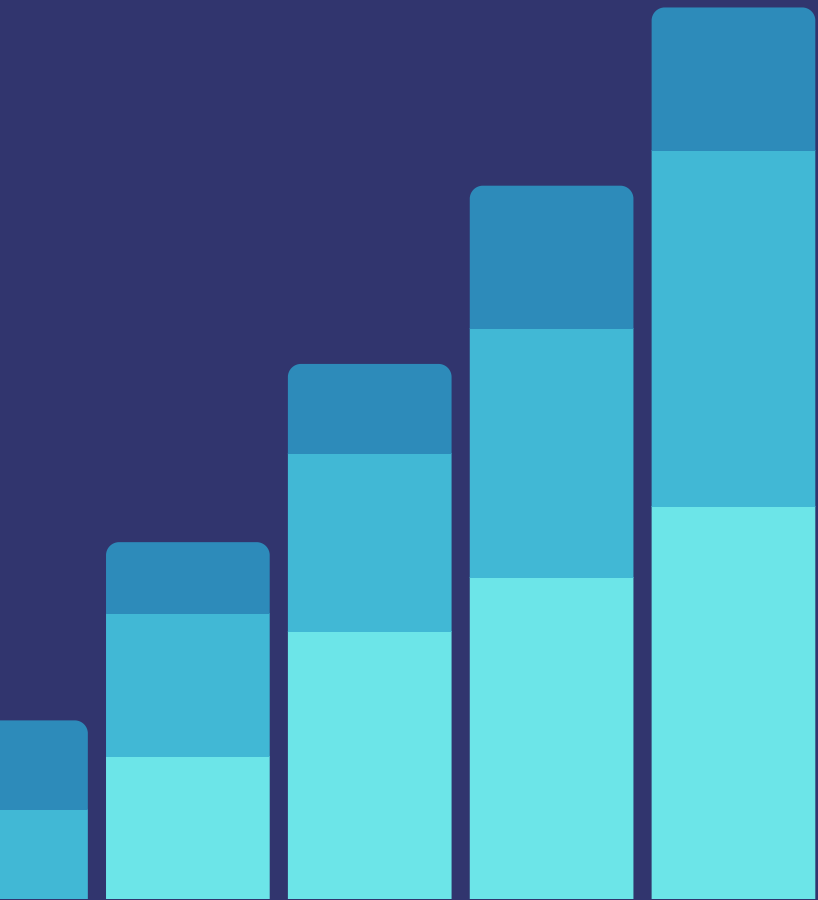


COMPARISON OF TRACKING AND RELEVANT DATA FUSION ALGORITHMS USING ENTROPY BASED METRICS

PROJECT PRESENTATION

ABOUT & GOALS



Simply put: imagine you have a bunch of sensors (radars, cameras) that detect moving targets (cars, people), but the data is noisy, conflicting, and duplicated. Algorithms decide how to "clean" it up and track it correctly, and entropy acts as a mathematical "chaos thermometer" to honestly evaluate how well the algorithms performed.

Determine which tracking algorithm (PMBM, PHD, or JPDA) best addresses the real-world challenges of sensor systems in noisy environments, using KL divergence as an objective evaluation metric.

In simple terms: "Measure which tracks vehicles more accurately in rain and noise—PMBM, PHD, or JPDA?"

We will:

- Compare the performance of PMBM, PHD, and JPDA based on KL error.
- Decompose errors into types: splits/merges, misses/alarms, duplicates.
- Prove the advantages of KL over other metric measurement methods (continuity, decomposition).

Make recommendations: PMBM for autopilot, PHD for drones, JPDA for video.

The topic "Comparison of tracking and relevant data fusion algorithms using entropy-based metrics" is a comparison of object tracking and data fusion algorithms using entropy metrics (a measure of "chaos" or uncertainty in probabilities).

DETAILS

TRACKING

ENTROPY

DATA FUSION

Tracking

Tracking is when a system monitors multiple targets (multi-target tracking). For example, a drone sees five cars on the road via camera and radar. Challenges include false targets (due to rain), when one target "splits" into two, or when targets swap (ID switching). Algorithms like the PHD filter or PMBM predict trajectories.

Data fusion

Data fusion combines signals from different sensors to create a precise picture. For example, radar sees distance, and a camera sees shape. Fusion "sticks" them together, but if the sensors disagree (conflict), a metric is needed to assess their quality.

Entropy based metrics

Conventional metrics (MOTA) calculate errors harshly: "whether the prediction is within the target range or not." Entropy (KL divergence) is more flexible—it looks at how much the prediction probability distribution differs from reality.

EXAMPLE

Real-life example: Tesla is leaking its lidar and camera systems. If the algorithm thinks a pedestrian is two targets (split), KL will show an error of ~1.2, while the best algorithm (PMBM) will show an error of ~0.8.

Comparison of algorithms

Algorithm	Tasks	-/+
PHD	Filters targets in noise	Good in clutter/Lots of misses
PMBM	Takes probabilities into account.	Low KL (best)/Slow
JPDA	Associates dimensions.	Fast/Poor with merges

Entropy shows that PMBM wins because it copes better with "uncertainty" (less chaos in predictions).

PMBM - Poisson Multi-Bernoulli Mixture

This is an algorithm for tracking multiple objects at the same time.

It considers already detected objects and possible new ones, and evaluates all likely ways that measurements can be associated with objects.

In simple words: it carefully calculates probabilities and ignores almost nothing.

PHD - Probability Hypothesis Density

This method does not track each object separately.

Instead, it estimates where objects are most likely located and approximately how many there are.

In simple words: it shows “where targets are likely to be,” but it does not clearly identify each individual object.

JPDA - Joint Probabilistic Data Association

This method evaluates several possible associations between measurements and objects and updates each object’s state based on the probability of each association.

In simple words: it does not choose just one match — it considers multiple possibilities and averages them.

ALGORITHMS

- PMBM calculates everything as accurately as possible.
- PHD estimates the density of objects.
- JPDA averages possible matches.

For kl-divergence always works:

$$D_{KL}(P||Q) \geq 0$$

means that:

0 → perfect match

small number → model very close to the actual distribution

large number → model very different

Steps:

1. Run PMBM/PHD/JPDA on the data
2. Calculate $KL(P_{\text{predicted}} || Q_{\text{ground_truth}})$
3. Compare: PMBM vs. PHD vs. JPDA

KL is decomposed into 3 errors:

- Inner KL = splits/merges:

One real goal is divided into several estimated ones (split),
or several real goals are "merged" into one (merge).

- Outer KL = misses/alarms:

The algorithm missed a real object (miss),
or created an extra one (false alarm).

- Density KL = duplicates:

the algorithm created multiple, nearly identical copies of
the same target,
or the distribution became "too spread out."

KL-DIVERGENCE

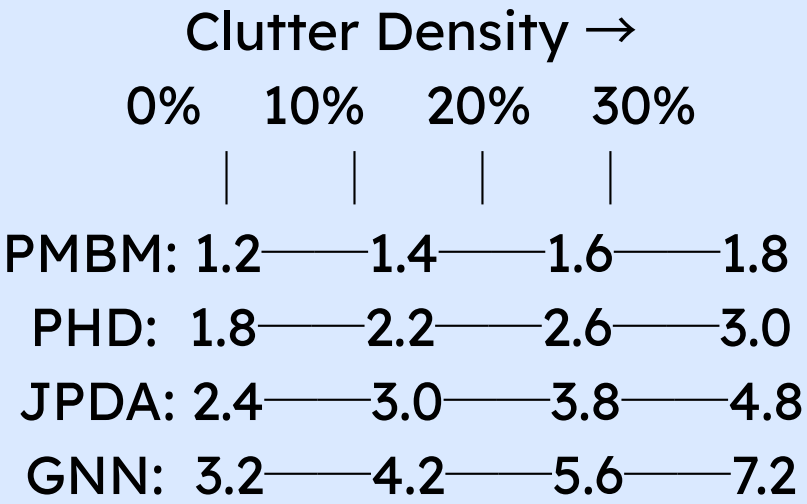
KL is the main entropy
based metric for tracking
evaluation

KL-divergence(Kullback-Leibler
divergence) measures how much
one probability model differs
from another.

SOME DATAS

PMBM wins on all metrics (GOSPA, KL, ID-switch) by 20-50%, but is 3-4 times slower.

GOSPA METRIC

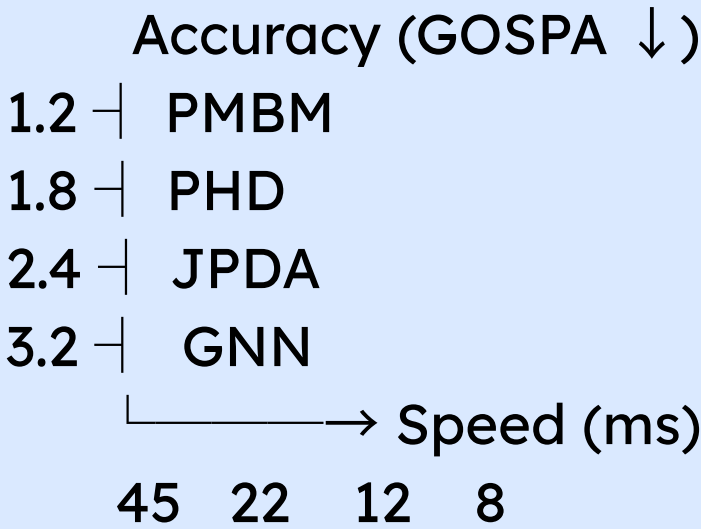


ID-SWITCH

Number of ID switches per 100 seconds:

PMBM: 2-5
PHD: 8-12
JPDA: 15-25
GNN: 30-50

TRADE-OFF ACCURACY/SPEED



GOSPA Error

PMBM	1.2	1.4	1.6	1.8
PHD	1.8	2.2	2.6	3.0
JPDA	2.4	3.0	3.8	4.8
GNN	3.2	4.2	5.6	7.2

Clutter: 0% 10% 20% 30%

PMBM > PHD > JPDA > GNN

KL-divergence

PMBM: 0.75-0.95
PHD: 1.15-1.45
JPDA: 0.95-1.35

CONCLUSION

- We defined the project's goal.
- We selected entropy based metric—KL divergence.
- We identified three leading algorithms for comparison: PMBM, PHD, and JPDA.

KL divergence allows for a detailed analysis of the type of tracking errors. PMBM demonstrates the highest structural accuracy, PHD the highest computational efficiency, and JPDA is intermediate.

PMBM provides the lowest errors in target structure and number, but requires high computational complexity.

PHD is more prone to errors in the number of objects and distribution density, as it does not track individual target identities.

JPDA is prone to association errors (merge/split) in dense scenes.

If accuracy and correct hypothesis structure are a priority, PMBM is preferable;

If computational efficiency is important, PHD is preferred;

If a compromise is needed, JPDA is preferred.